Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiment 8

Attention Mechanisms in ResNet50 for Flower Image Classification

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

1. Objective

The objective is to implement and compare attention mechanisms within an ImageNet-pretrained ResNet50 classifier for five-class flower recognition. The study examines attention type, insertion stage, multiple insertion positions, channel/spatial order, spatial-attention kernel size, and additional mechanisms. Performance is assessed using classification metrics, model complexity, inference latency, training curves, confusion matrices, and qualitative visualizations.

2. Learning Outcomes

The experiment studies: (i) channel, spatial, self, combined, SE, ECA and CBAM attention; (ii) stage-wise attention placement; (iii) the effect of applying attention at multiple stages; (iv) channel–spatial ordering; (v) spatial-kernel size; (vi) transfer learning and fine-tuning; (vii) accuracy, precision, recall, macro F1 and confusion-matrix analysis; and (viii) the accuracy–complexity and F1–latency trade-offs.

3. Dataset and Experimental Setup

Dataset: TensorFlow Flowers ('tf_flowers'‘). The task is five-class image classification with the classes dandelion, daisy, tulips, sunflowers and roses. The notebook resizes images to 224×224 pixels and uses the ResNet50 preprocessing pipeline. The data split is 70% training, 15% validation and 15% testing. The recorded test set contains 550 images. A fixed seed of 42 is used in the notebook.

Backbone	ResNet50, ImageNet-pretrained
Input size	$224 \times 224 \times 3$
Batch size	8
Feature-extraction epochs	3
Fine-tuning epochs	2
Dropout	0.3
Feature-extraction learning rate	10^{-3}
Fine-tuning learning rate	As configured in notebook
Loss	Categorical cross-entropy
Test images	550

The epoch counts are the values used in this run; they are not presented as mandatory defaults from the manual. The same split and common training procedure are used for comparisons where applicable.

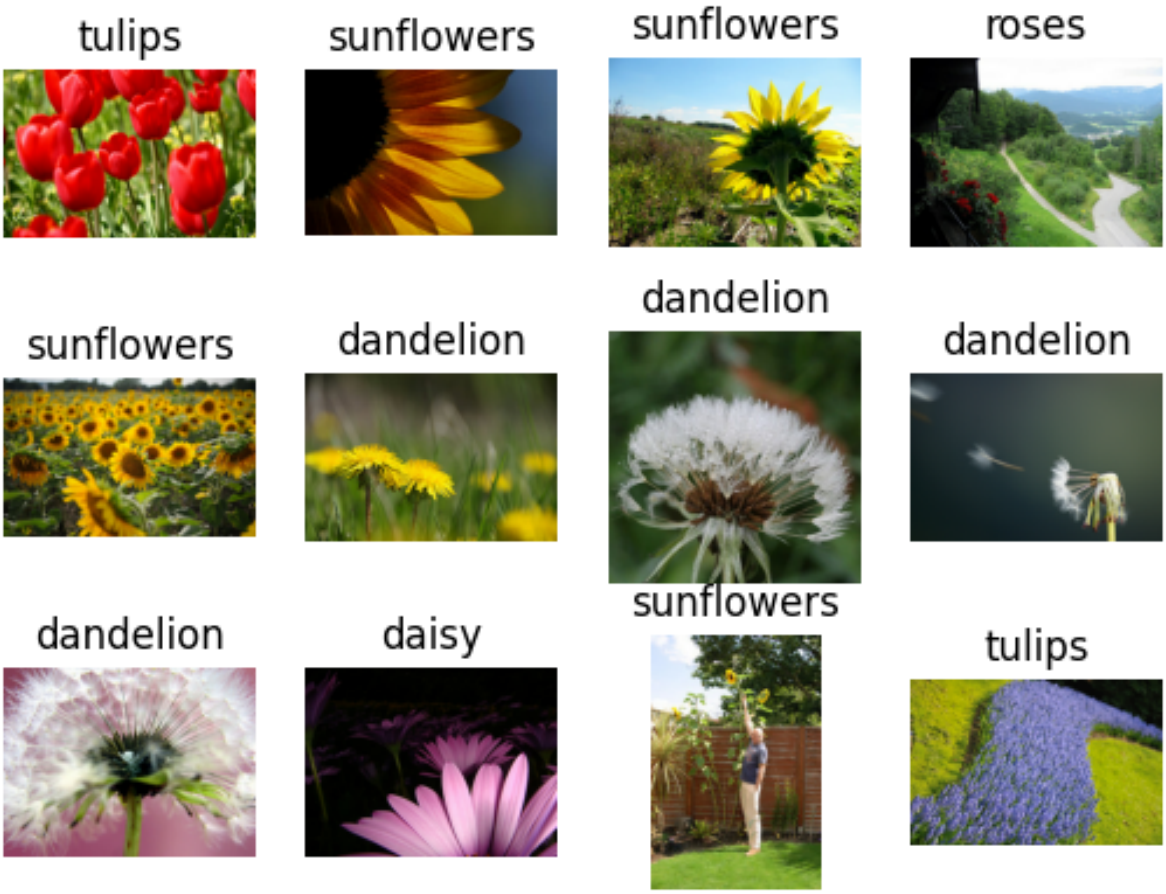


Figure 1: Example images from the dataset.

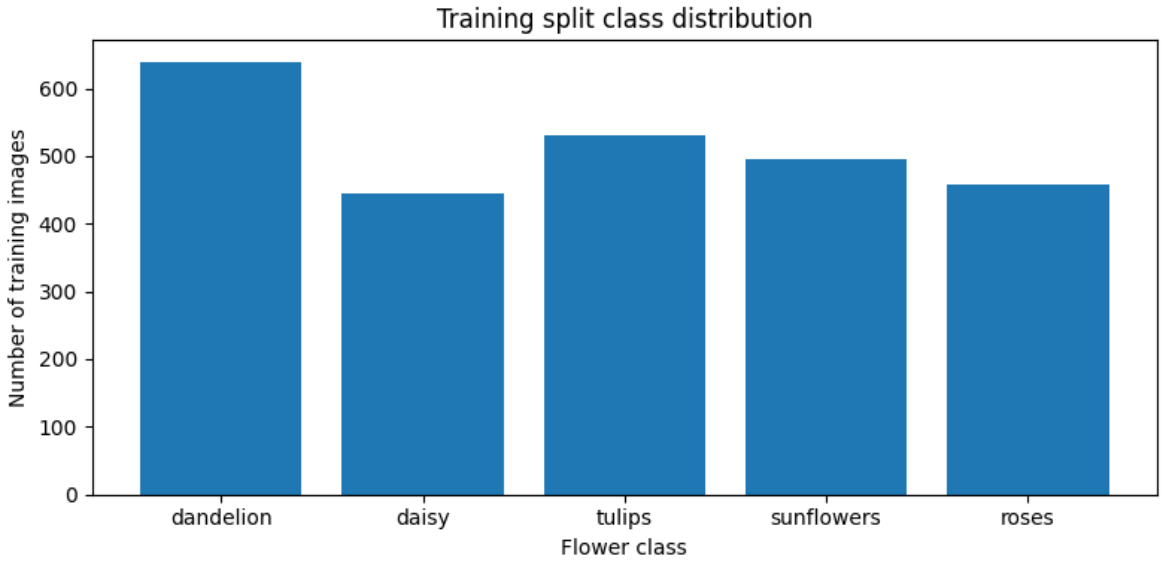
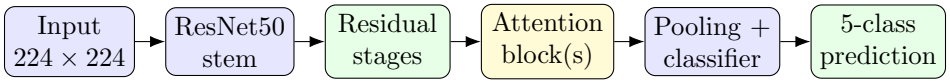


Figure 2: Class distribution recorded by the notebook.

4. Model and Attention Overview

ResNet50 extracts progressively higher-level features through residual stages. Attention blocks reweight feature channels, spatial positions, or relationships between spatial locations. Channel attention learns which feature channels are useful; spatial attention learns where informative features occur; self-attention models interactions between positions; SE and ECA are channel-focused mechanisms; CBAM combines channel and spatial attention.



5. Training Curves

The notebook records three feature-extraction epochs followed by two fine-tuning epochs. The curves below are taken from the executed notebook outputs. They show the evolution of training and validation metrics; short runs mean the curves should be interpreted as evidence of this run only, not as a full convergence study.

5.1 Baseline ResNet50

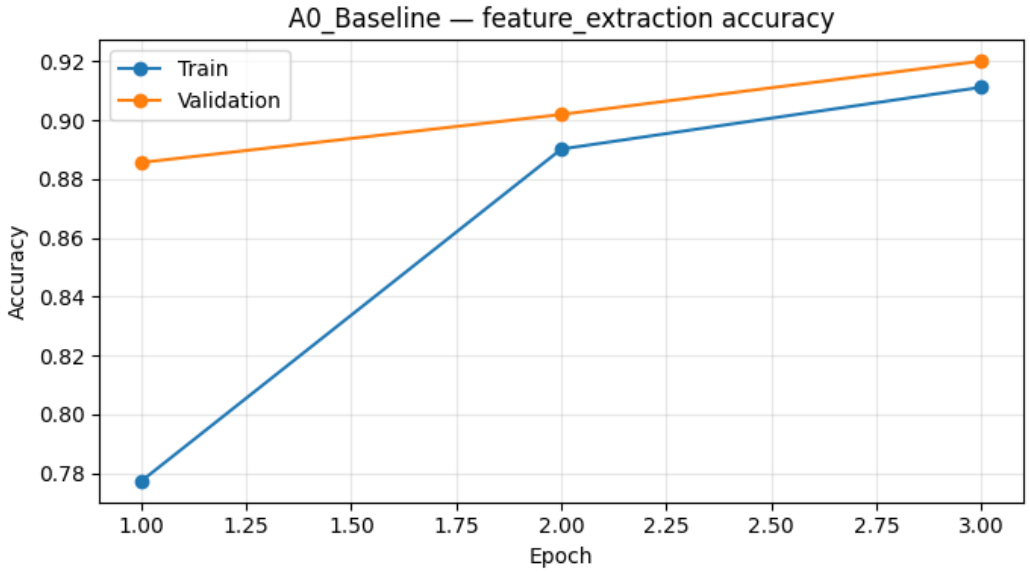


Figure 3: Baseline feature-extraction accuracy.

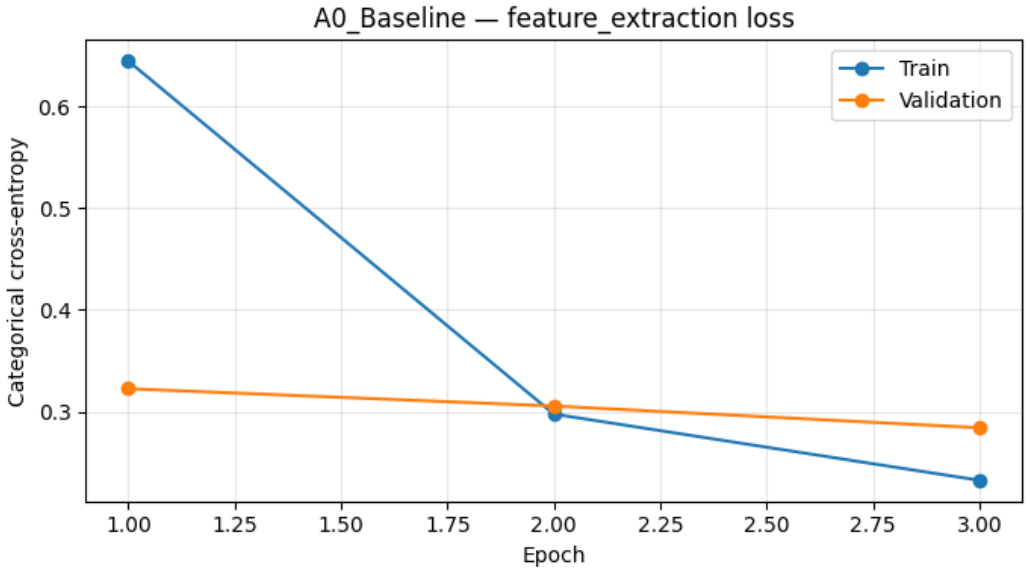


Figure 4: Baseline feature-extraction loss.

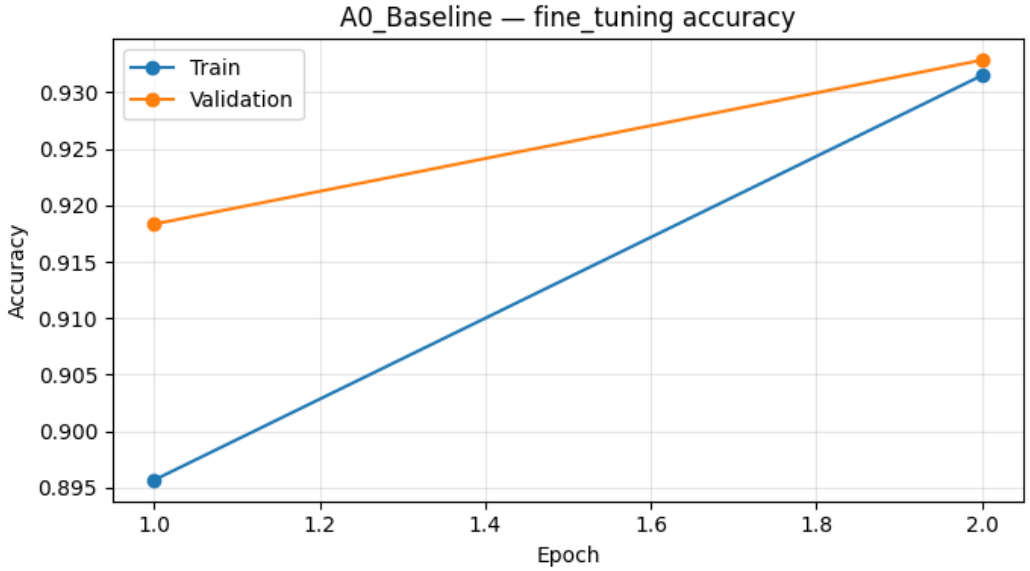


Figure 5: Baseline fine-tuning accuracy.

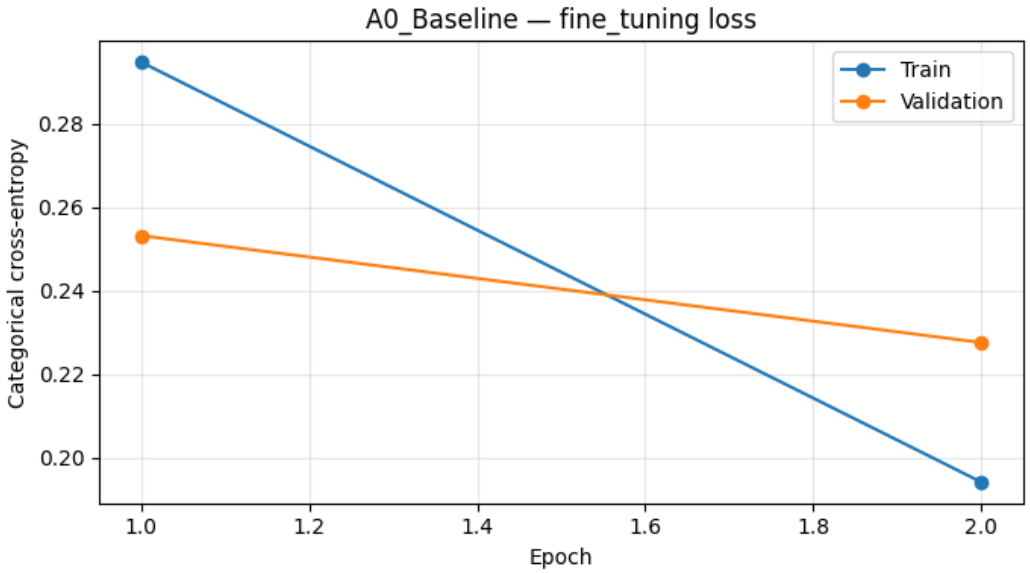


Figure 6: Baseline fine-tuning loss.

5.2 Manual Experiment 2: Attention after conv4_x

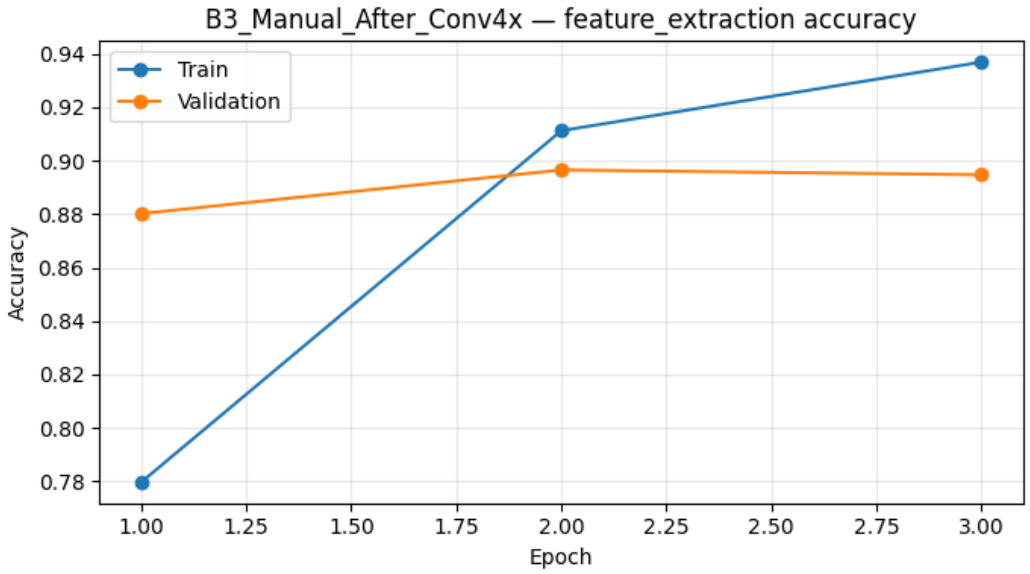


Figure 7: Feature-extraction accuracy for channel–spatial attention after conv4_x.

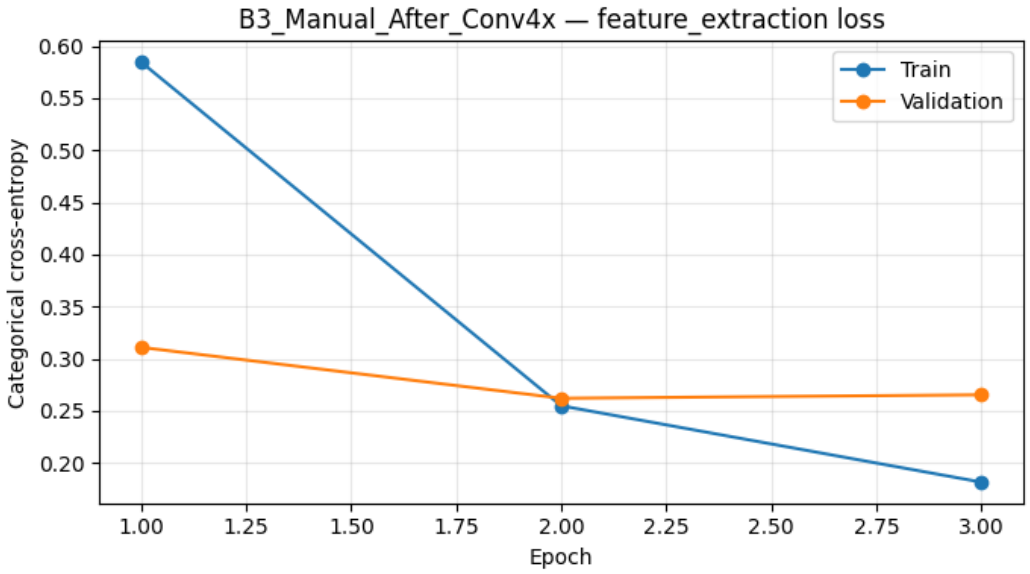


Figure 8: Feature-extraction loss for channel–spatial attention after conv4_x.

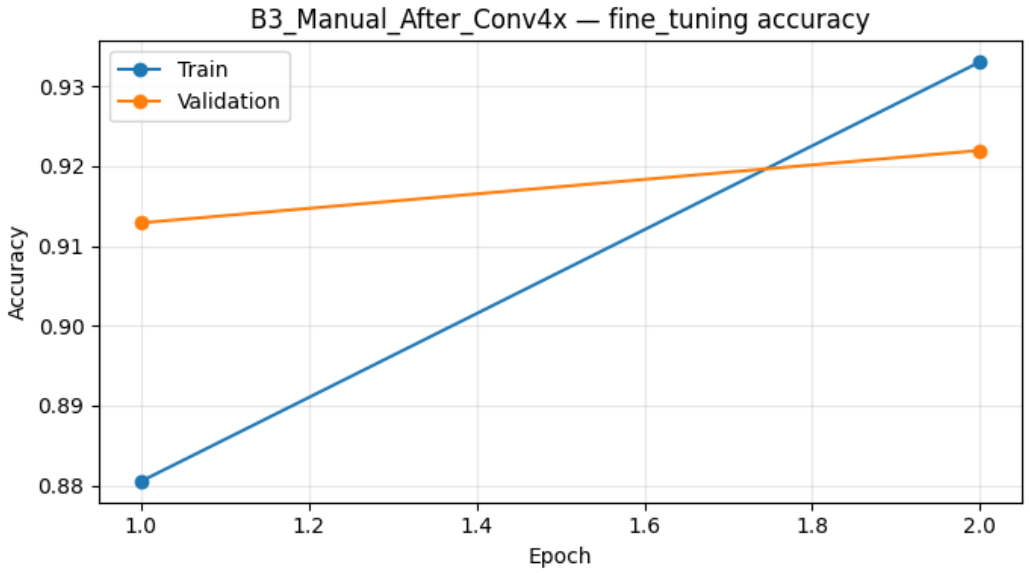


Figure 9: Fine-tuning accuracy for channel–spatial attention after conv4_x.

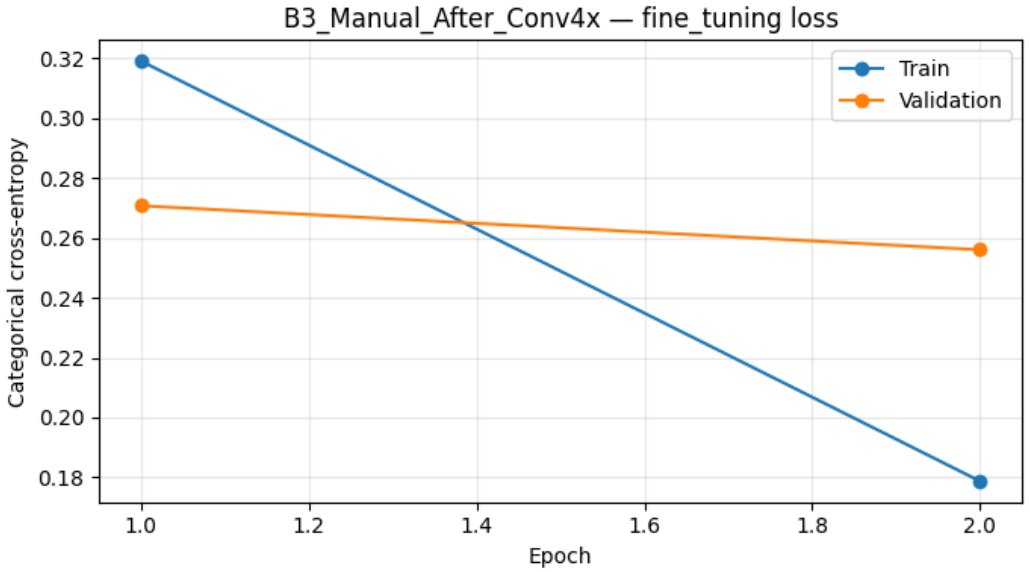


Figure 10: Fine-tuning loss for channel–spatial attention after conv4_x.

6. Main Experimental Results

The experiment headings below follow the Lab 8 manual's experiment numbering (manual Sections 14–19).

Experiment 1 (Manual Section 14): Attention-Type Ablation

The baseline was compared with channel, spatial, channel–spatial, self-attention and combined attention at P3 (after conv4_x). The baseline and spatial-attention P3 models both recorded 93.28% final validation accuracy. The complete per-model results are provided in Section 7 so that lower-performing variants are also represented. The experiment demonstrates that adding an attention module does not guarantee a higher score under a fixed short training budget.

Experiment 2 (Manual Section 15): Attention-Position Ablation

The same channel–spatial attention design was trained after conv2_x, conv3_x, conv4_x and conv5_x. The feature resolutions/channels specified by the manual are 56×56/256, 28×28/512, 14×14/1024 and 7×7/2048, respectively. The recorded validation accuracies for the manual-aligned variants were 89.29% (conv2_x), 90.93% (conv3_x), 92.20%

(conv4_x) and 92.92% (conv5_x). On the held-out test set, the conv4_x variant obtained 93.27% accuracy and macro F1 of 93.29%, the highest test accuracy among the evaluated variants in the recorded table. These results are specific to this run.

Experiment 3 (Manual Section 16): Multiple Attention Positions

The notebook compares attention at P3 only, P4 only and P3+P4. Their recorded final validation accuracies are shown in the all-experiment table. The purpose is to test whether repeated attention yields enough predictive improvement to justify extra parameters and computation. In this run, P3+P4 does not outperform the best single-position configurations on validation or test metrics.

Experiment 4 (Manual Section 17): Attention Order

D1 applies channel attention followed by spatial attention; D2 reverses the order. D2 reaches 92.92% final validation accuracy and 92.91% test accuracy, while D1 reaches 92.20% final validation accuracy and 91.27% test accuracy. Under this run's settings, spatial–channel ordering performed better, although repeated seeds would be needed to establish whether the difference is robust.

Experiment 5 (Manual Section 18): Spatial-Attention Kernel Size

Spatial attention with 3×3, 5×5 and 7×7 kernels was evaluated at P3. Their recorded final validation accuracies were 90.56%, 92.01% and 92.01%, respectively. The 5×5 and 7×7 variants were close in validation accuracy; the held-out test table should be considered separately rather than assuming the best validation result also has the best test result.

Additional Task (Manual Section 19): Additional Attention Mechanisms

The notebook includes SE, ECA and CBAM at P3, satisfying the requirement to compare at least two additional mechanisms. Their test metrics, model parameters and latency are listed in Section 7. SE is competitive with the baseline, while ECA and CBAM show slightly lower recorded test accuracy in this run. The differences are small and should not be generalized beyond the single experimental run.

7. Quantitative Comparison

Table 1 combines validation accuracy, held-out test accuracy, test macro precision, recall and macro F1, total parameters and recorded inference latency. The test metrics are generated from the notebook's report-ready comparison cell. Missing values are not imputed.

Experiment	Val. acc. (%)	Test acc. (%)	Precision (%)	Recall (%)	Macro F1 (%)	Parameters	ms/image
A0 Baseline	93.28	92.36	92.57	92.47	92.50	23597957	29.91
A2 Spatial P3	93.28	91.64	91.93	91.62	91.74	23598055	29.44
A3 Channel+Spatial P3	93.10	91.64	91.87	91.81	91.81	23730215	30.20
A5 Combined P3	92.92	92.91	93.05	93.00	93.01	24254503	30.37
D2 Spatial-to-Channel	92.92	92.91	93.45	92.79	93.01	23730215	29.66
E Kernel 7x7	92.92	92.18	92.13	92.48	92.24	23598055	29.17
Exp.2 B4 after conv5_x	92.92	91.82	91.94	91.72	91.76	24124519	29.58
C2 P4 only	92.74	92.73	92.88	92.69	92.74	24124519	29.67
C1 P3 only	92.74	92.55	92.77	92.71	92.70	23730215	29.51
A1 Channel P3	92.56	92.36	92.53	92.40	92.46	23730117	29.67
A6 SE P3	92.56	92.36	92.68	92.28	92.39	23730117	29.77
Exp.2 B3 after conv4_x	92.20	93.27	93.19	93.40	93.29	23730215	29.42
B1 Channel+Spatial P2	92.20	90.91	91.05	91.06	91.02	23631367	29.77
D1 Channel-to-Spatial	92.20	91.27	91.56	91.31	91.38	23730215	29.83
A7 ECA P3	92.01	92.00	91.90	92.06	91.95	23597960	29.92
E Kernel 5x5	92.01	92.36	92.43	92.42	92.41	23598007	29.34
C3 P3+P4	92.01	91.64	91.72	91.72	91.70	24256777	30.55
B2 Channel+Spatial P3	91.65	92.18	92.55	92.12	92.28	23730215	30.04
A8 CBAM P3	90.93	91.82	91.95	91.91	91.92	23730215	29.86
Exp.2 B2 after conv3_x	90.93	90.73	91.12	90.98	91.00	23631367	29.54
E Kernel 3x3	90.56	90.55	90.70	90.54	90.58	23597975	29.37
A4 Self-attention P3	90.38	90.36	90.50	90.67	90.53	24122245	29.74
Exp.2 B1 after conv2_x	89.29	87.45	87.59	87.63	87.59	23606519	29.82

Table 1: Recorded validation and held-out test results. Latency is environment-dependent.

The highest held-out test accuracy in the executed comparison is Exp.2 B3 after conv4_x at 93.27%, with macro F1 93.29%. The baseline test accuracy is 92.36%. The selected model variable in the notebook's earlier evaluation cell resolved to A0_Baseline; therefore, the report does not treat the baseline confusion matrix as a confusion matrix for the highest-test-accuracy model. This distinction is retained to avoid mislabelling an output.

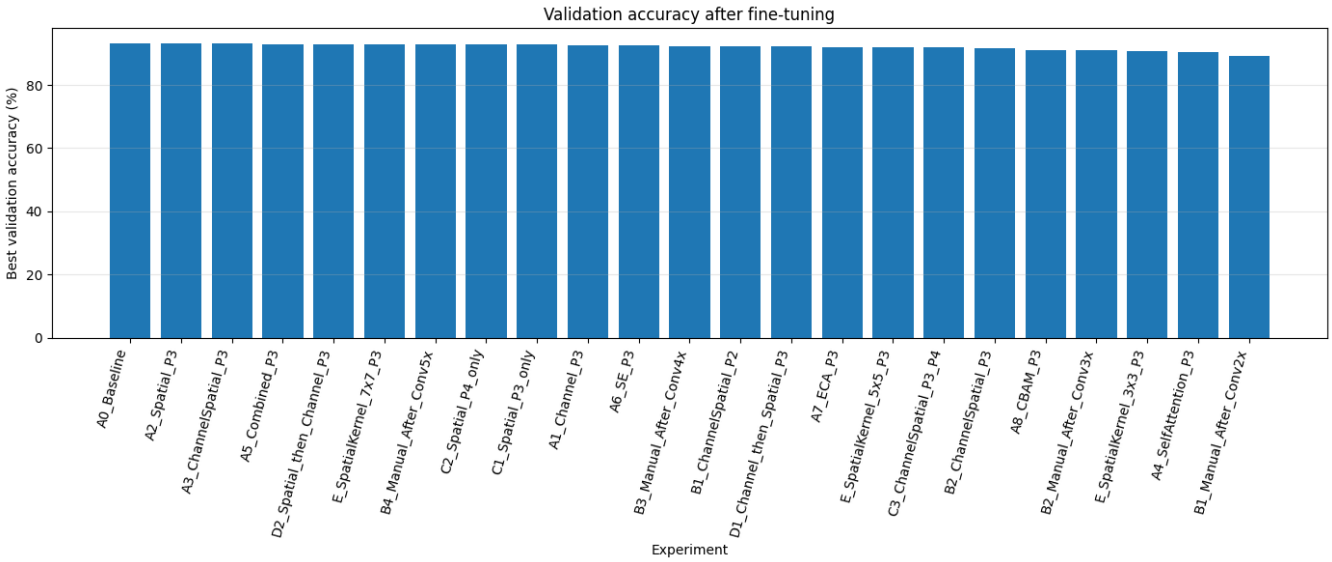


Figure 11: Fine-tuning validation-accuracy comparison across completed experiments.

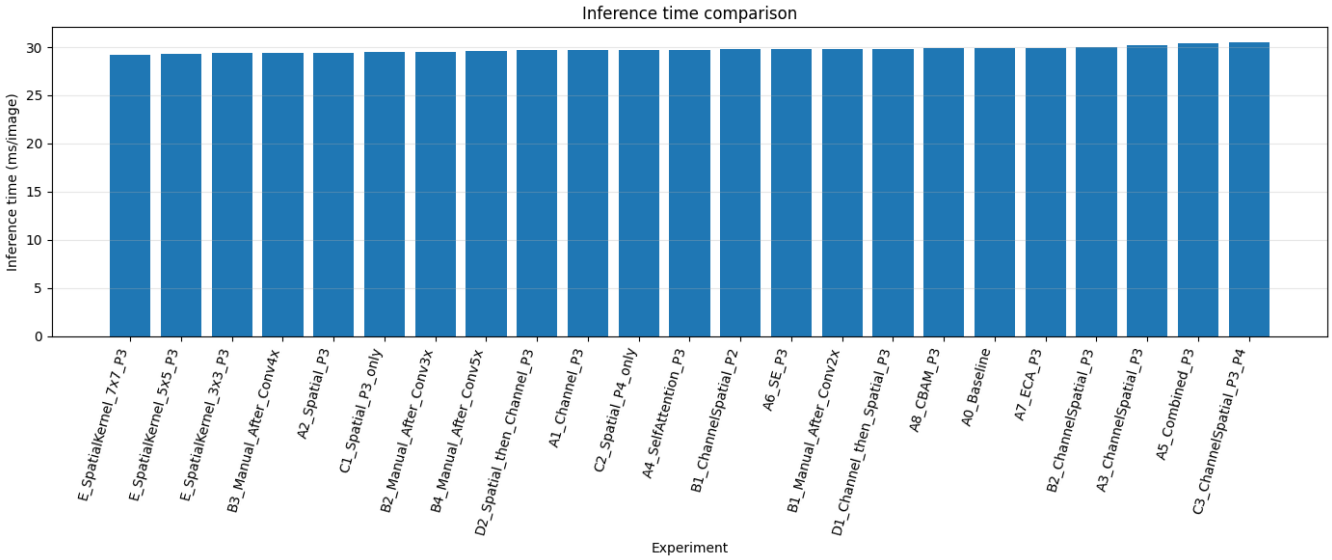


Figure 12: Recorded inference-time comparison across experiments.

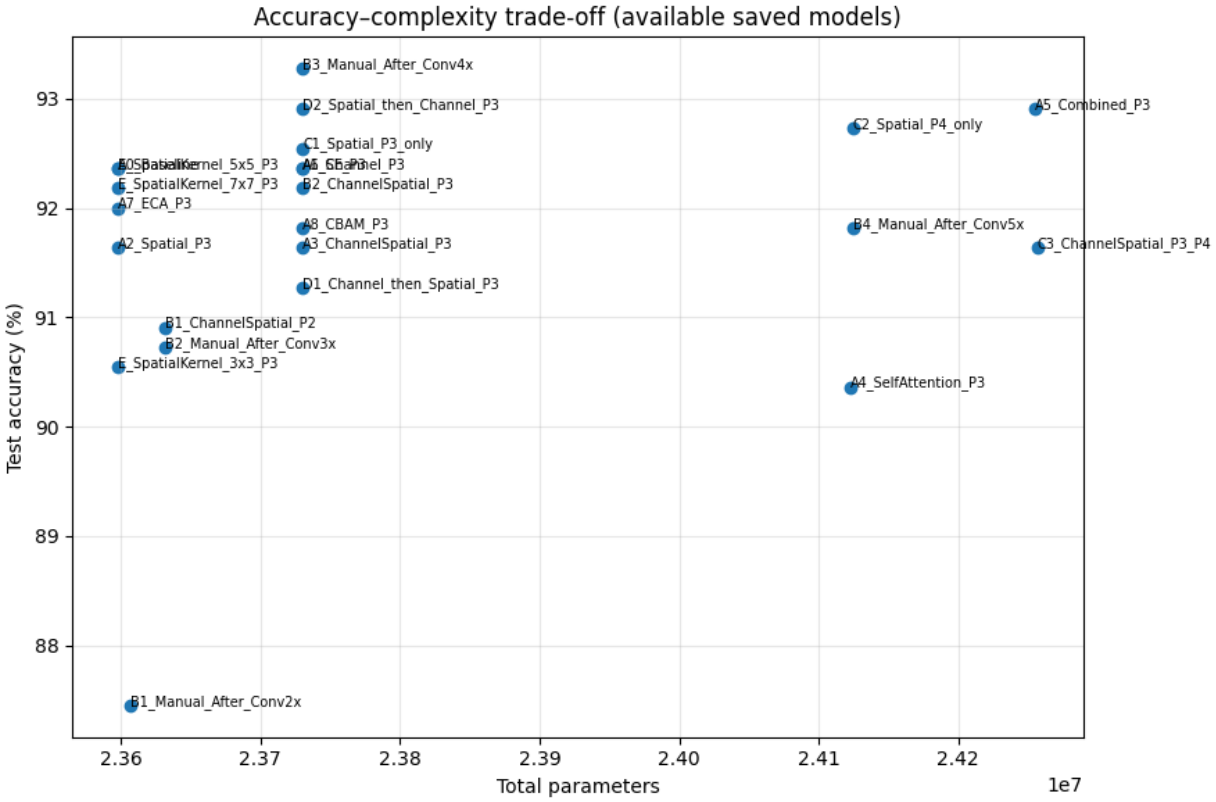


Figure 13: Accuracy–model-complexity trade-off.

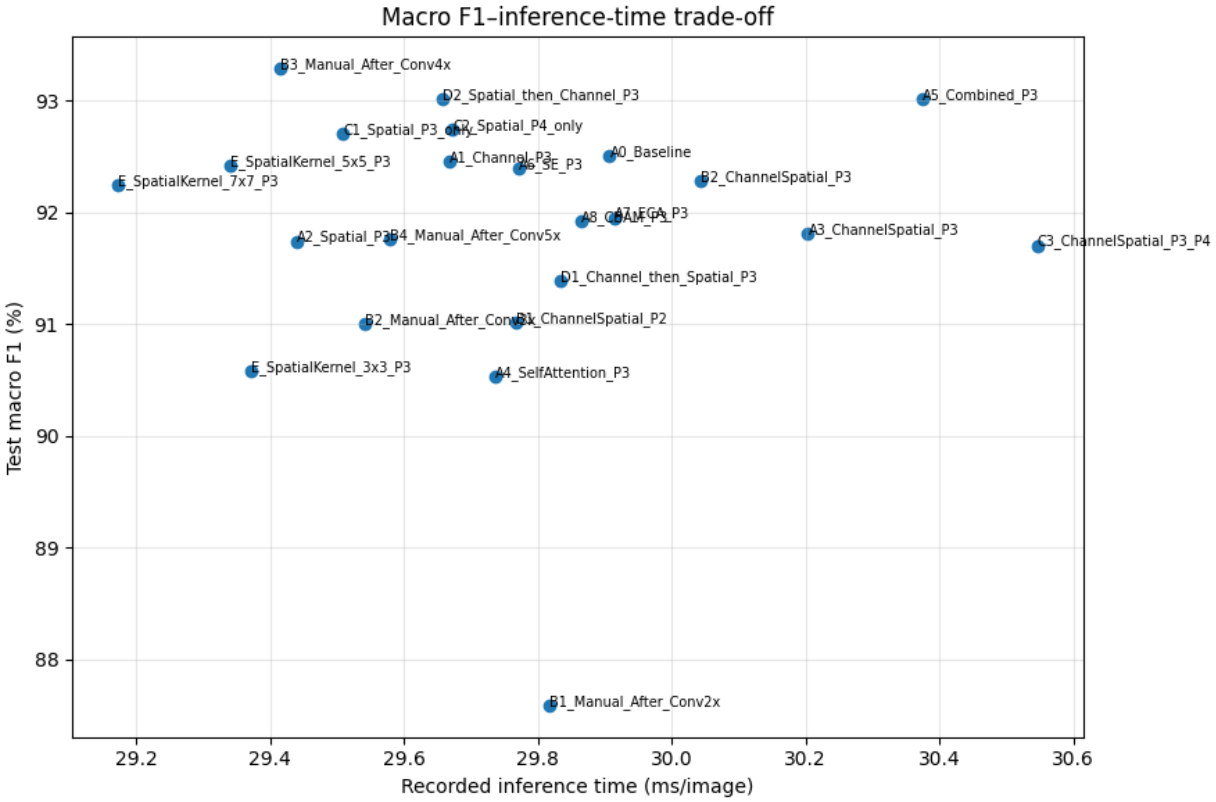


Figure 14: Macro-F1 versus inference-latency trade-off.

8. Confusion Matrix and Class-wise Analysis

The executed notebook contains a baseline confusion matrix and a class-wise classification report for the model referenced by its evaluation cell. Because that cell's selected-model name resolved to A0_Baseline, the class-wise metrics below are reported as baseline metrics rather than being attributed to the highest-test-accuracy B3 manual-position model.

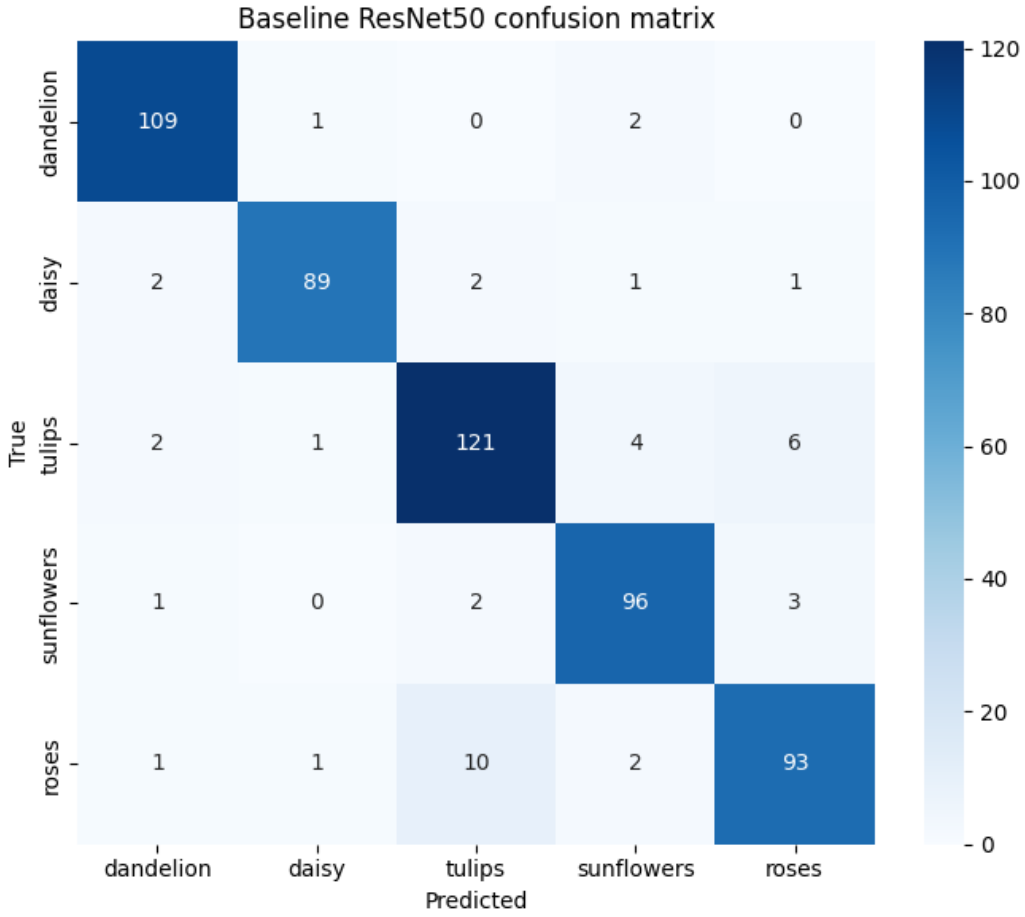


Figure 15: Confusion matrix for the baseline ResNet50.

Class	Precision	Recall	F1-score	Support
Dandelion	0.95	0.97	0.96	112
Daisy	0.97	0.94	0.95	95
Tulips	0.90	0.90	0.90	134
Sunflowers	0.91	0.94	0.93	102
Roses	0.90	0.87	0.89	107

The baseline report shows the lowest recall for roses (0.87), while dandelions have the highest recall (0.97). A confusion matrix reveals which classes are confused with each other, information that a single accuracy score cannot provide.

9. Attention and Feature Visualizations

9.1 Feature Maps Before and After Attention

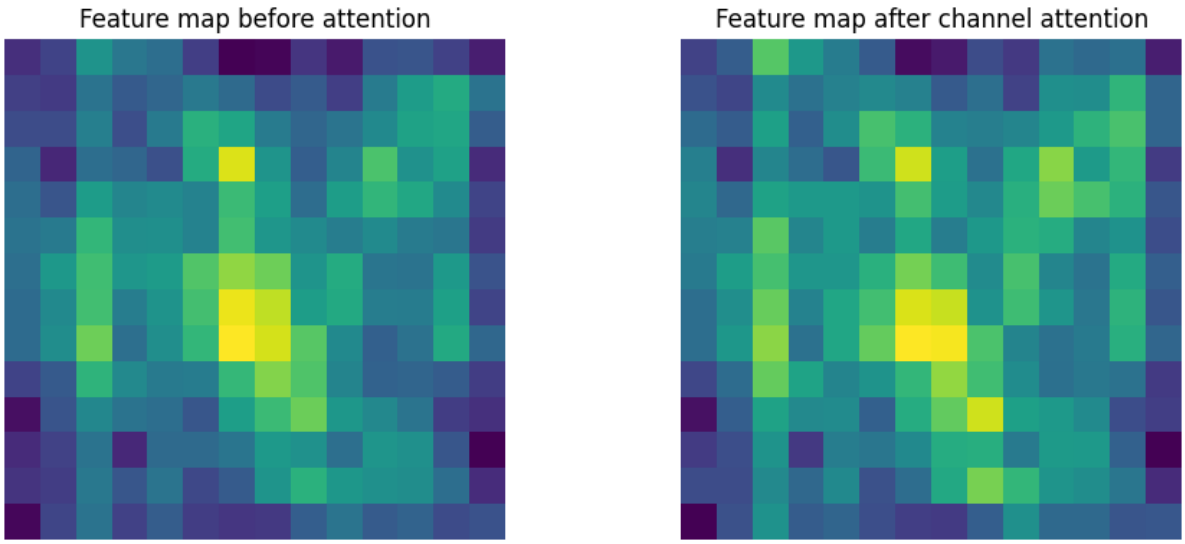


Figure 16: Feature-map response before and after channel attention.

9.2 Channel Attention

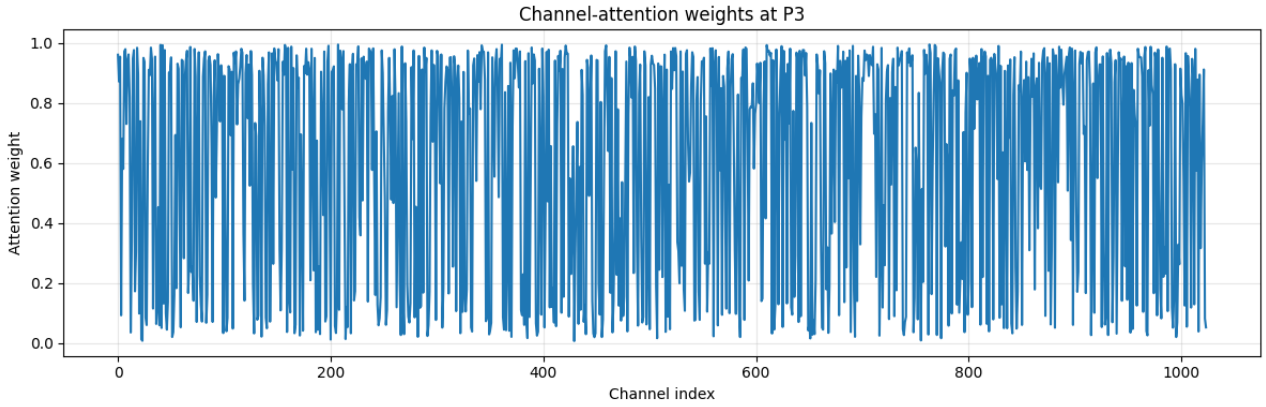


Figure 17: Channel-attention weights.

9.3 Spatial Attention

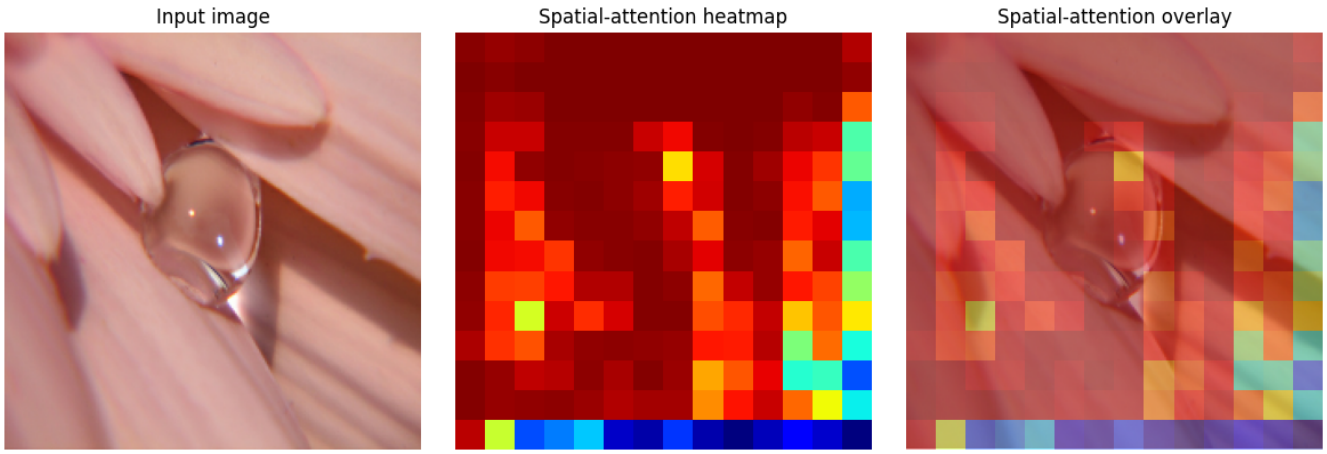


Figure 18: Spatial-attention map and overlay.

9.4 Self-Attention Response

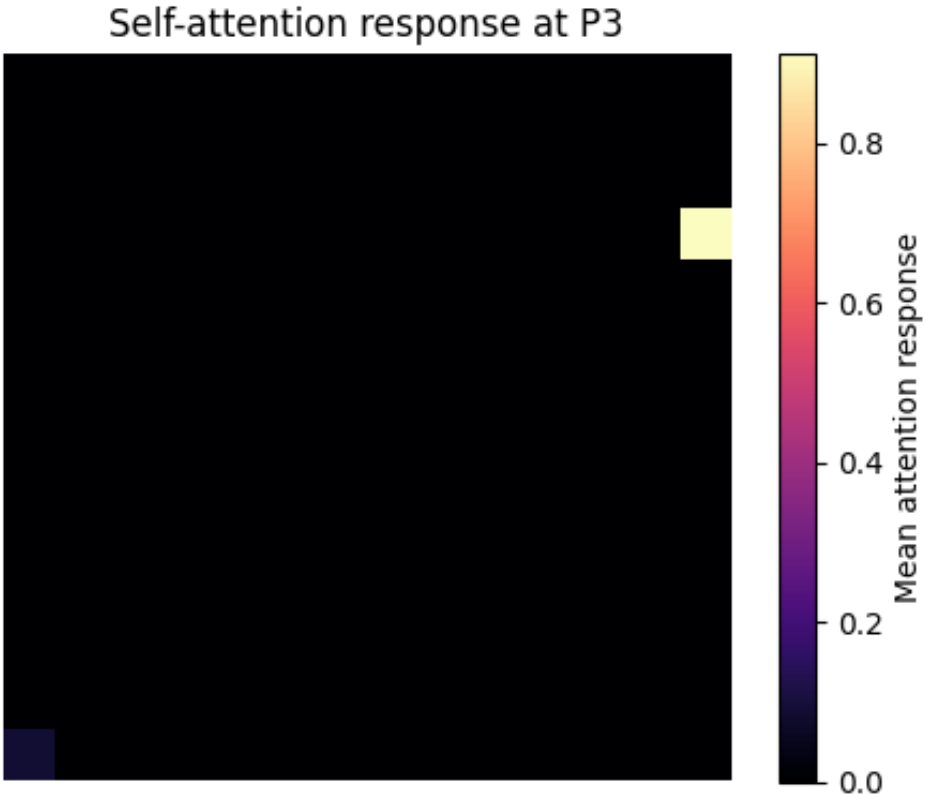


Figure 19: Self-attention response.

9.5 Grad-CAM

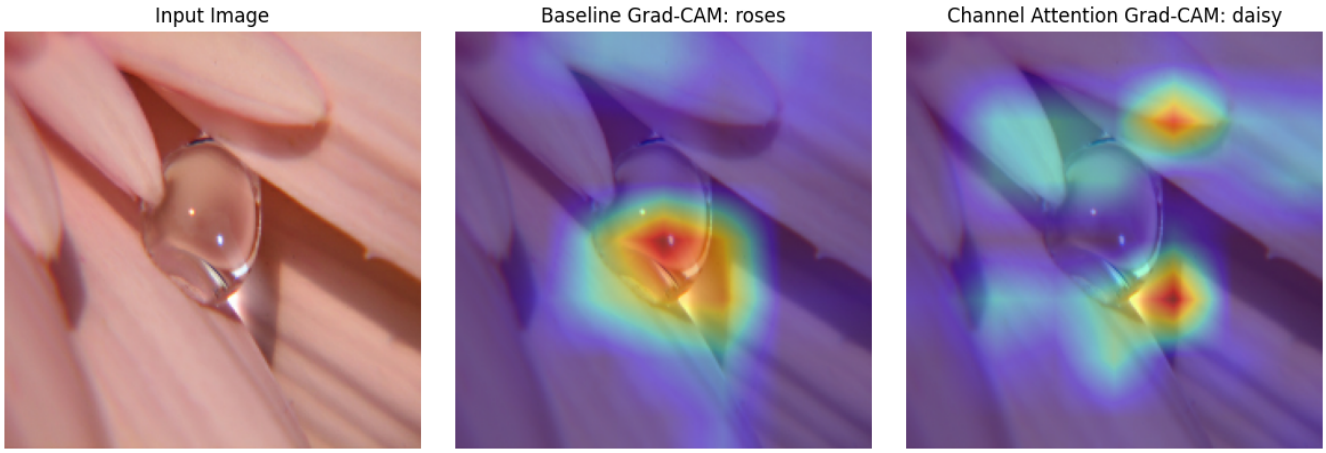


Figure 20: Baseline and channel-attention Grad-CAM comparison.

Grad-CAM is a post-hoc visualization of class-score gradients with respect to convolutional feature maps. It is distinct from an internal attention map and should be interpreted as a qualitative explanation rather than proof of causal importance.

9.6 Attention Position: P2, P3 and P4

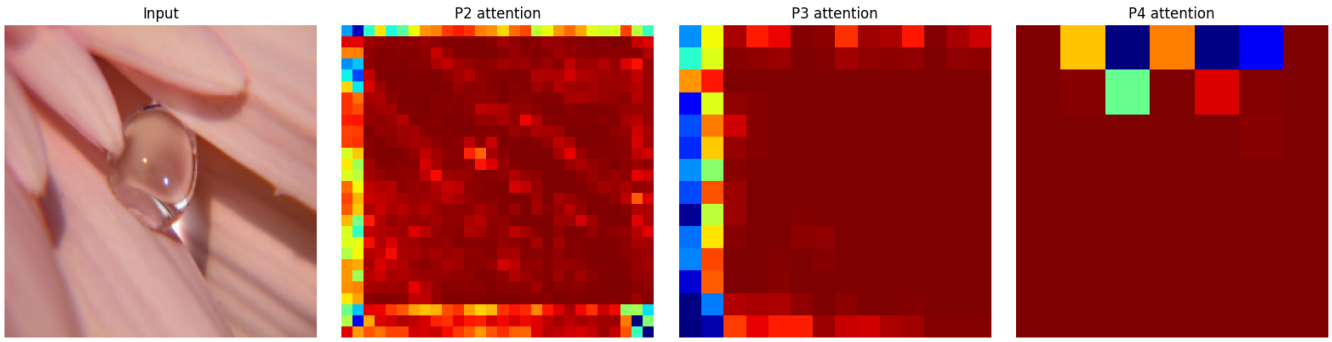


Figure 21: Attention maps at P2, P3 and P4.

10. Qualitative Error Inspection

The notebook produced a five-image comparison with three correctly classified and two incorrectly classified examples. However, because the selected-model variable resolved to the baseline, the executed comparison used the baseline as both comparison targets. This figure is therefore included only as an example-selection record and is not used to claim that attention improves localization or error handling.

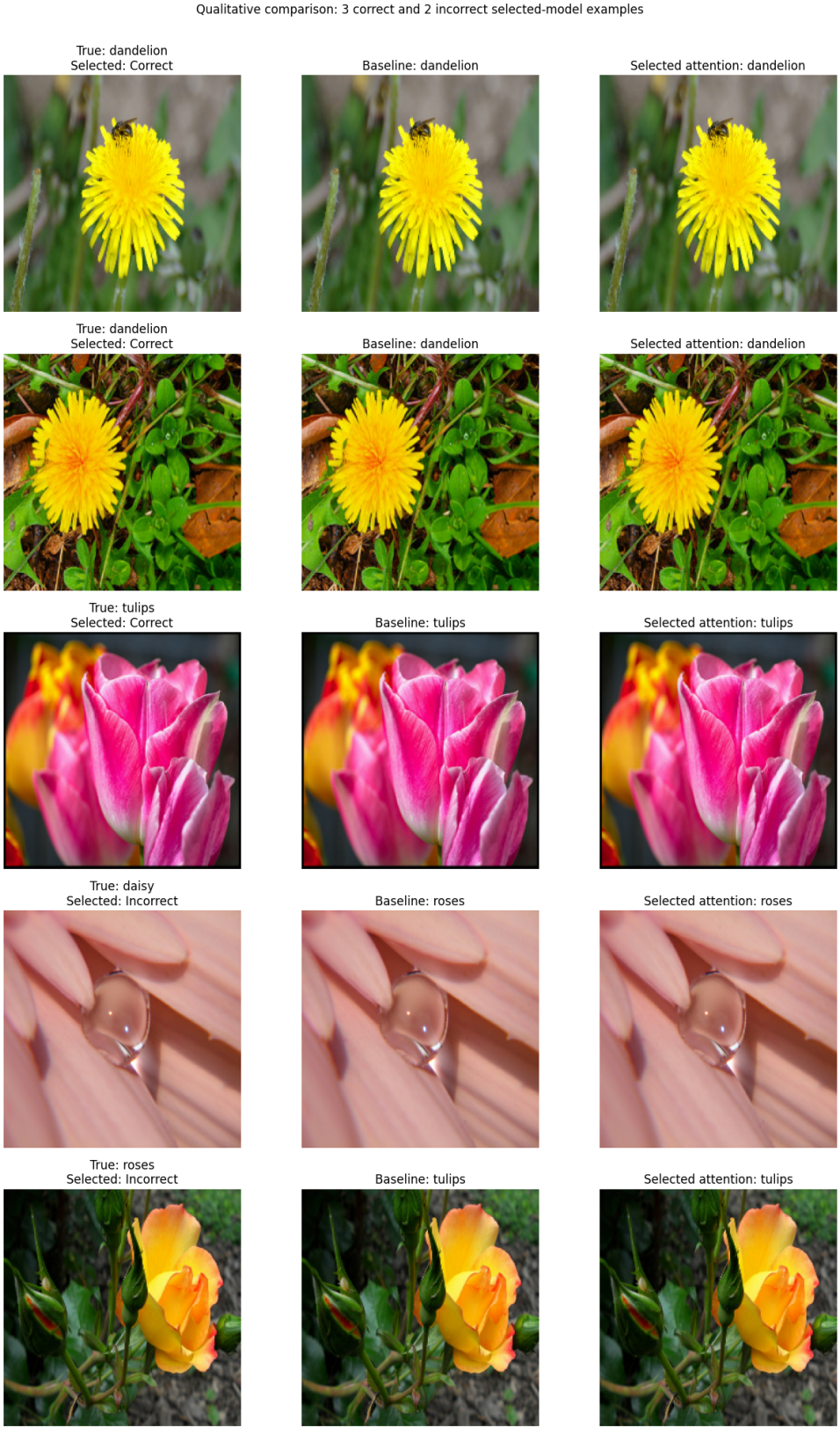


Figure 22: Three correct and two incorrect examples recorded by the notebook. The comparison target resolves to the baseline in this execution.

A valid across-mechanism qualitative conclusion requires rerunning the evaluation-only cell after explicitly selecting the intended attention model and its matching attention map and weights. This is an evaluation fix, not a need to retrain models.

11. Discussion

The experiments show that the effect of attention depends on its type and insertion point. The manual-aligned conv4_x channel–spatial model has the highest observed held-out test accuracy (93.27%) and macro F1 (93.29%) in the recorded comparison. D2 spatial–channel ordering is also competitive. Multiple attention positions do not automatically improve results, and additional parameter count alone is not a reliable indicator of predictive quality. The inference-time measurements are clustered around approximately 29–31 ms per image in this runtime; such measurements depend on hardware, warm-up and batch/synchronization conditions.

The main limitations are the short training schedule (three feature-extraction epochs and two fine-tuning epochs), a single recorded run, environment-dependent inference timing, and the selected-model variable mismatch in the qualitative/confusion-matrix evaluation. Attention maps and Grad-CAM are qualitative aids; they do not independently establish causal explanations. The report therefore separates recorded validation results from held-out test metrics and does not attribute baseline-only plots to a different model.

12. Conclusion

This experiment implemented a ResNet50 transfer-learning pipeline and compared attention type, position, multiple positions, channel–spatial order, kernel size, and SE/ECA/CBAM mechanisms on five flower classes. Across the evaluated models, channel–spatial attention after conv4_x achieved the highest recorded test accuracy of 93.27% and macro F1 of 93.29%. The baseline achieved 92.36% test accuracy, showing a modest improvement for the strongest test-scoring variant in this run. Further repeated-seed experiments and a corrected evaluation-only selection cell are recommended before making strong general claims.

13. References

1. K. He, X. Zhang, S. Ren and J. Sun, "Deep Residual Learning for Image Recognition," CVPR, 2016.
2. J. Hu, L. Shen and G. Sun, "Squeeze-and-Excitation Networks," CVPR, 2018.
3. S. Woo et al., "CBAM: Convolutional Block Attention Module," ECCV, 2018.
4. Q. Wang et al., "ECA-Net: Efficient Channel Attention for Deep Convolutional Neural Networks," CVPR, 2020.
5. TensorFlow Datasets, "tf_flowers," `https://www.tensorflow.org/datasets/catalog/tf_flowers`.
6. TensorFlow/Keras Applications, ResNet50 documentation, `https://keras.io/api/applications/resnet/`.

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